Online Appendix

Anatomy of a Smooth Transition: A Task-Based Macroeconomic Model of AGI Displacement with Fiscal and Social Safety Nets

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C External empirical anchoring of the social-instability parameters

The social-instability channel contains three reduced-form parameters: the social-crash cutoff ς^{th} (baseline 0.50), the stress coefficient α_s (baseline 0.60), and the sustained-exceedance window J (baseline 10 periods, one model period being one year). We search the empirical literature for mappable evidence for each, distinguish direct anchors from indirect directional evidence, and state the mapping assumptions explicitly. Table 1 summarizes the anchoring status.

Table 1: External empirical anchoring status of the three social-instability parameters.

Parameter	Baseline	Anchoring status	Nature of evidence
ς^{th}	0.50	no direct external anchor	directional/continuous probability only; no normalized cutoff
α_s	0.60	partially anchored	illustrative reverse-engineered range $\approx$ 1.0–6.6; baseline 0.60 is below this range, a conservative setting
J	10	partially anchored	mean/upper-tail scale (crisis 8–10 yrs; civil-war mean 11.1 yrs)

C.1 The cutoff $\varsigma^{th} = 0.50$: no direct external anchor. No study provides a population-dependency threshold that maps onto a 0.50 cutoff of the $[0, 1]$ normalized stress variable. The authoritative literature consistently reports continuous probability or dose–response relationships rather than structural thresholds replicable across samples. Using rainfall as an instrument across 41 African countries over 1981–1999, Miguel, Satyanath and Sergenti [1] identify that a five-percentage-point drop in real GDP growth raises the probability of conflict in the following year (at least 25 battle deaths) by about 12 percentage points, a relative increase of roughly 50%. Ponticelli and Voth [2], across 25 European countries over 1919–2008, find that a one-standard-deviation spending cut (about 2.8 percentage points of GDP) adds

roughly 0.4 unrest events per year, with the event count nearly doubling as cuts widen from 1% to 10% of GDP; only a descriptive soft cutoff around a 2%-of-GDP cut appears, not an identified structural threshold. Using natural disasters as instruments, Bellemare [3] identifies that a one-unit rise in the global food price index adds about 0.99 food-related unrest reports per month, while price volatility has no significant effect. Earlier work by Collier and Hoeffler [13], Fearon and Laitin [14], and Alesina and Perotti [15] likewise reports continuous marginal effects. The only explicit numerical cutoff, due to Lagi, Bertrand and Bar-Yam (2011, arXiv:1108.2455, not peer reviewed), clusters protests when the FAO food price index exceeds 210; but that cutoff is a price-index level rather than a population dependency ratio, and mapping it onto $\varsigma^{th} = 0.50$ would require an over-long chain of normalization assumptions, so we do not adopt it as a direct anchor.

C.2 The stress coefficient $\alpha_s = 0.60$: partially anchored, with an identifiable range. The sign with which unemployment or relief dependence raises property crime and social unrest is robust across U.S., European and cross-country panels, ruling out $\alpha_s \leq 0$. The literature reports crime semi-elasticities to unemployment of roughly +1.3% to +4% per percentage point of unemployment, concentrated in property crime with weak effects on violent crime. Raphael and Winter-Ebmer [4] find that a one-percentage-point fall in unemployment lowers property crime by about 1.6%–2.4%; Lin [5] identifies a 1.8% OLS and 4.0% 2SLS rise in property crime per extra percentage point of unemployment; Gould, Weinberg and Mustard [6] find that the 23.3% decline in low-skill male wages over 1979–1993 accounts for 43% of the rise in property crime; Buonanno, Drago and Galbiati [7], across about 20 developed countries over 1979–2010, identify a roughly 1.3% rise in total crime and 1.7% rise in homicides per extra percentage point of unemployment; and Fougère, Kramarz and Pouget [8] identify a significant positive effect of youth unemployment on property crime in France; see also the surveys by Freeman [16] and Draca and Machin [17]. Mapping these semi-elasticities $\beta_{crime} \approx \partial \log C / \partial u$ onto $\alpha_s = \partial \,\text{stress} / \partial r$ requires three layers of assumptions, and we keep every quantity in consistent units: (A) the normalized range linking the stress index to disorder indicators, setting stress = 1 to correspond to a 3–10-fold amplification of crime over baseline (mapping slopes f' of about 0.43–0.91); (B) the conversion factor $k = r/u$ between the relief share r and unemployment u; and (C) a local linearization treating the semi-elasticity as approximately constant over the working range. On (B), rather than import an external value, we use the model's own accounting: relief recipients are the unemployed not absorbed by public employment, $r = (1 - p_E)u$, so at the baseline $p_E = 0.45$ the conversion is $k = 1 - p_E = 0.55$ (verified directly from the simulated paths). On units, when u is measured on the $[0, 1]$ scale the semi-elasticity entering the chain is $\beta_{crime} = 1.3$–4 (the per-percentage-point 1.3%–4% figures restated per unit of the fraction u). Substituting into $\alpha_s \approx f' \beta_{crime} (1/k)$ with $k = 0.55$ gives an illustrative reverse-engineered interval of roughly 1.0–6.6 (the lower value $0.43 \times 1.3/0.55 \approx 1.02$, the upper $0.91 \times 4/0.55 \approx 6.62$). We read this as a wide, normalization-dependent illustration rather than an empirical estimate: the interval spans more than a factor of six and moves almost one-for-one with the assumed stress-to-disorder mapping. The baseline 0.60 lies below even the lower end of this interval, so the calibration deliberately errs on the conservative side, assigning a smaller stress response than the reverse engineering would tolerate; it is a scenario setting rather than a value the

interval identifies.

C.3 The sustained-exceedance window $J = 10$: partially anchored at mean/upper-tail scale. No study directly estimates the window of consecutive stress exceedance that triggers a crash. Annual-scale evidence brackets ten years along two independent lines. First, after systemic banking crises, per-capita GDP returns to its pre-crisis peak in about eight years on average (median 6.5 years), with 45% of episodes experiencing a second dip [9]; Cerra and Saxena [10] find highly persistent, near-permanent output losses after banking crises, with later estimates placing the average post-crisis GDP loss at about ten years. Second, civil wars last 11.1 years on average (median 7.1 years) [11], and the hazard estimates of Collier, Hoeffler and Söderbom [12] average just over seven years, with wars begun in the 1980s lasting 105 months (about 8.75 years). On the other side, protest waves and crime shocks persist only months to 2–5 years, markedly shorter than ten years. $J = 10$ is therefore a defensible mean/upper-tail anchor aligned with chronic post-crisis slumps and average civil wars, but it lies above the median durations and should be read as such rather than as a median anchor.

C.4 Parameter decision and empirical-range robustness. Because none of the three parameters has a precise, directly identified empirical replacement, we keep the baseline values but state their standing accurately: assigning a value to ς^{th} would lack mappable threshold evidence; $\alpha_s = 0.60$ is a conservative setting that sits below the ≈ 1.0–6.6 reverse-engineered interval rather than inside it; and $J = 10$ is a supported mean/upper-tail value. An empirical-range sweep extends the α_s scan across the full 0.30–6.62 interval alongside a directional ς^{th} scan. The analytic boundaries are $\alpha_s^{crit} = 0.50/((1 - 0.45)u^{max}) = 0.909$ (with $u^{max} = 1$) and $\varsigma_{crit}^{th} = 0.33$, and the numerical sweep locates the first social crash at $\alpha_s = 0.91$ (smooth at 0.90) and between $\varsigma^{th} = 0.30$ and 0.33 (smooth at 0.33); $\min C$ stays 0.357 throughout because the social channel does not feed back into the economic paths. Two results bound the channel. First, the ranking is robust across the entire interval: the pure transfer crashes at every α_s, whereas the corner stays smooth through $\alpha_s = 0.90$. Second, the corner's own smoothness is not unconditional: under the upper portion of the reverse-engineered interval ($\alpha_s \geq 0.91$), the corner also triggers the social channel. The sweep therefore reports both the conservative margin at the baseline and the parameter region in which even the workfare corner would fail, rather than presenting the reduced-form channel as settled.

D Extended robustness runs

This section collects the three extensions summarized in the body: the functional-form comparison of the share law, the capability-sensitivity sweep, and the counterfactual that removes the logistic erosion term. All runs are reproduced by the named analysis scripts.

D.1 Functional form of the share law. The equation for the endogenous rate is a modelling choice, so we re-solve the corner with the probit share, $\omega_t = \Phi(\ell_0 + \Gamma_t - \beta_A g_t)$ where Φ is the standard normal cumulative distribution. Because reusing the logistic intercept would make the baseline probability 0.0016 rather than 0.05, the intercept is re-calibrated to the

normal quantile $\Phi^{-1}(0.05) = -1.645$, so both links start from the same baseline $\omega_0 = 0.05$ and the comparison is like for like. Measured over the policy-driven horizon $t \geq 1$, the effective capture peaks at 0.121 under the logistic and 0.239 under the probit. The consumption buffer is effectively unchanged, $\min_t C_t$ moving only from 0.3562 to 0.3556. The logistic corner completes absorption (smooth), whereas the probit corner keeps the buffer but does not finish absorbing within $T = 80$, a right-censored not-reached outcome rather than a crash. The link function therefore does not move the consumption floor or trigger a crash, which remain governed by skill depreciation, but it does change the time needed to finish absorption. This is reproduced by `analysis/check_link_function.py`.

D.2 Capability-sensitivity sweep. On the $T = 200$ horizon, varying $\beta_A \in \{0, 1, 3\}$ separates a mid-horizon crossing from a durable reinstatement; the critical value is 0.5667. At $\beta_A = 0$, with no capability erosion, the strong floor crosses at $t = 29$ and the effective capture peaks at 0.731 and stays above the threshold, so the crossing is durable in the sense that it is not reversed; the market employment share ends the horizon at a positive 0.344 without sliding back, although it still declines at a small negative rate rather than settling. At $\beta_A = 1$ the strong floor crosses at $t = 35$, with a peak of 0.640 above the critical value, but continued erosion pulls the effective capture back below the threshold later, so the employment share keeps declining to 0.172: a mid-horizon crossing that defends consumption and employment over the short run without a durable reinstatement. At $\beta_A = 3$ the peak falls to 0.486, below the critical value, there is no crossing and the employment share reaches zero. The corner remains smooth throughout and its minimum consumption stays in 0.356–0.357, while the strong floor's minimum consumption stays in 0.433–0.456 at every reported β_A, so the fiscal buffer defends consumption at every setting even though a durable restoration of the labor engine occurs only in the no-erosion limit. The baseline $\beta_A = 1$ is the conservative middle setting, not an estimated value.

D.3 Counterfactual: the collapse is not built into the logistic term. The law contains a monotone term $-\beta_A g_t$ with $g_t = \log(\text{cap}_t/A_0)$ accumulating over the horizon, so a reader could ask whether the labor-share collapse is imposed by the functional form rather than generated by the model. The direct counterfactual sets $\beta_A = 0$, which removes the erosion channel and leaves only the policy stock and the task accounting, and re-runs the same engine with every other parameter fixed. The no-policy path still crashes, with $\min C = 0.0575$ and the employment share absorbed to zero, and the financed corner still stays smooth with $\min C = 0.356$. The collapse therefore follows from the task-accounting condition $\omega\sigma\, dN < \kappa_I A$ of Theorem 1 in the main manuscript, which holds even when the reinstatement rate is not being eroded, rather than from the monotone logistic term. The β_A coefficient instead governs the durability of a strong-policy reinstatement, the terminal share 0.344 versus 0.172 reported above, not whether the no-policy path crashes. This counterfactual is reproduced by `analysis/check_capability_counterfactual.py`.

References

[1] E. Miguel, S. Satyanath and E. Sergenti, Economic Shocks and Civil Conflict: An Instrumental Variables Approach, *Journal of Political Economy* 112(4), 725–753, 2004, DOI: 10.1086/421174.

[2] J. Ponticelli and H.-J. Voth, Austerity and Anarchy: Budget Cuts and Social Unrest in Europe, 1919–2008, *Journal of Comparative Economics* 48(1), 1–19, 2020, DOI: 10.1016/j.jce.2019.09.007.

[3] M. F. Bellemare, Rising Food Prices, Food Price Volatility, and Social Unrest, *American Journal of Agricultural Economics* 97(1), 1–21, 2015, DOI: 10.1093/ajae/aau038.

[4] S. Raphael and R. Winter-Ebmer, Identifying the Effect of Unemployment on Crime, *Journal of Law and Economics* 44(1), 259–283, 2001, DOI: 10.1086/320275.

[5] M.-J. Lin, Does Unemployment Increase Crime? Evidence from U.S. Data 1974–2000, *Journal of Human Resources* 43(2), 413–436, 2008, DOI: 10.3368/jhr.43.2.413.

[6] E. D. Gould, B. A. Weinberg and D. B. Mustard, Crime Rates and Local Labor Market Opportunities in the United States: 1979–1997, *Review of Economics and Statistics* 84(1), 45–61, 2002, DOI: 10.1162/003465302317331919.

[7] P. Buonanno, F. Drago and R. Galbiati, Response of Crime to Unemployment, *Journal of Contemporary Criminal Justice* 30(1), 29–40, 2014, DOI: 10.1177/1043986213509023.

[8] D. Fougère, F. Kramarz and J. Pouget, Youth Unemployment and Crime in France, *Journal of the European Economic Association* 7(5), 909–938, 2009, DOI: 10.1162/JEEA.2009.7.5.909.

[9] C. M. Reinhart and K. S. Rogoff, Recovery from Financial Crises: Evidence from 100 Episodes, *American Economic Review Papers and Proceedings* 104(5), 50–55, 2014 (NBER Working Paper 19823), DOI: 10.3386/w19823.

[10] V. Cerra and S. C. Saxena, Growth Dynamics: The Myth of Economic Recovery, *American Economic Review* 98(1), 439–457, 2008, DOI: 10.1257/aer.98.1.439.

[11] J. D. Fearon, Why Do Some Civil Wars Last So Much Longer Than Others?, *Journal of Peace Research* 41(3), 275–301, 2004, DOI: 10.1177/0022343304043770.

[12] P. Collier, A. Hoeffler and M. Söderbom, On the Duration of Civil War, *Journal of Peace Research* 41(3), 253–273, 2004, DOI: 10.1177/0022343304043769.

[13] P. Collier and A. Hoeffler, Greed and Grievance in Civil War, *Oxford Economic Papers* 56(4), 563–595, 2004, DOI: 10.1093/oep/gpf064.

[14] J. D. Fearon and D. D. Laitin, Ethnicity, Insurgency, and Civil War, *American Political Science Review* 97(1), 75–90, 2003, DOI: 10.1017/S0003055403000534.

[15] A. Alesina and R. Perotti, Income Distribution, Political Instability, and Investment, *European Economic Review* 40(6), 1203–1228, 1996, DOI: 10.1016/0014-2921(95)00030-5.

[16] R. B. Freeman, The Economics of Crime, in O. Ashenfelter and D. Card (eds.), *Handbook of Labor Economics*, vol. 3, ch. 52, Elsevier, Amsterdam, 3529–3571, 1999.

[17] M. Draca and S. Machin, Crime and Economic Incentives, *Annual Review of Economics* 7, 389–408, 2015, DOI: 10.1146/annurev-economics-080614-115808.